

Air Quality Prediction

Problem Statement

Air pollution has become one of the leading environmental challenges, affecting human health and contributing to climate change. Poor air quality increases the risk of respiratory diseases, cardiovascular problems, and environmental degradation. Monitoring air quality manually is time-consuming, making it essential to develop an AI-based system that can predict pollution levels and support timely decision-making.

Project Overview

Our team developed **EcoAir AI**, a machine learning-based air quality prediction system that analyzes environmental sensor data to estimate **Carbon Monoxide (CO) concentration**. The project uses historical air quality data, performs preprocessing and visualization, and trains a **Random Forest Regression** model to provide accurate pollution predictions. The system demonstrates how Artificial Intelligence can contribute to environmental monitoring and sustainable development.

Solution Offered

- Data preprocessing and cleaning of air quality sensor readings.
- Exploratory Data Analysis (EDA) using various visualizations.
- Machine Learning model to predict **CO(GT)** concentration.
- Feature importance analysis to identify major pollution factors.

Who Are The End Users?

- Environmental monitoring agencies.
- Smart city administrators.
- Government pollution control boards.
- Researchers and students.
- Industries monitoring emissions.

Technology Used To Solve The Problem

Machine Learning & Data Science

- Python – Core programming language.
- Pandas – Data cleaning and preprocessing.
- NumPy – Numerical computations.
- Scikit-learn – Random Forest Regression model.

Data Visualization

- Matplotlib – Statistical charts.
- Seaborn – Heatmaps, scatter plots, pair plots, and feature analysis.

Dataset

- UCI Air Quality Dataset containing pollutant sensor readings such as CO, NO₂, NO_x, Temperature, Relative Humidity, and Absolute Humidity.